Markov Decision Processes

The MDP Setup for Reinforcement Learning

Roadmap: the one-line spine

A discounted MDP makes returns bounded; value / Q functions satisfy Bellman fixed-point equations; the Bellman operators are γ-contractions, so those fixed points exist uniquely and are reached geometrically.

model $\to$ objective $\to$ value functions $\to$ Bellman equations

$\to$ operators $\to$ contraction

$\to$ **VI / PI / GPI** (known model) $\to$ **SARSA / Q-learning** (unknown model)

Scope

Tabular MDPs and algorithms built directly on the Bellman operators. Function approximation, policy gradients, and full average-reward theory are out of scope. Proofs omitted here.

Probability prerequisites

P1. Random variables, expectation, distributions

Basics

- A random variable X takes values in a set with a probability law.
- Expectation: for finite-valued X, $\mathbb{E}[X] = \sum_x x \Pr(X = x)$.
- Linearity: $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$ — always, even if X, Y dependent.

Distributions over a finite set

A distribution on $\mathcal{S}$ is a vector $p \in \Delta(\mathcal{S})$, the probability simplex:

$$\Delta(\mathcal{S}) = \left\{ p \in \mathbb{R}^{\mathcal{S}} : \; p(s) \geq 0, \; \sum_s p(s) = 1 \right\}.$$

Used by: the transition kernel $P(\cdot \mid s, a) \in \Delta(\mathcal{S})$ and policy $\pi(\cdot \mid s) \in \Delta(\mathcal{A})$; the return G_t and value functions V^π, Q^π are expectations.

P2. Conditional expectation & the tower property

Conditional probability / expectation

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}, \qquad \mathbb{E}[X \mid Y = y] = \sum_x x \Pr(X = x \mid Y = y).$$

$\mathbb{E}[X \mid Y]$ is itself a random variable (a function of Y).

Law of total expectation (tower property)

$$\mathbb{E}[X] = \mathbb{E}\big[\mathbb{E}[X \mid Y]\big], \qquad \mathbb{E}[X \mid Z] = \mathbb{E}\big[\mathbb{E}[X \mid Y, Z] \,\big|\, Z\big].$$

Used by: every Bellman derivation. Conditioning $G_t = R_t + \gamma G_{t+1}$ on the first action and transition gives $V^\pi = \sum_a \pi(a \mid s) Q^\pi(s, a)$ and $Q^\pi(s, a) = r(s, a) + \gamma \sum_{s'} P(s' \mid s, a) V^\pi(s')$.

P3. Stochastic processes, Markov chains, estimators

Stochastic process & Markov property

A stochastic process is a sequence $S_0, S_1, \dots$ of random variables. It is a Markov chain if the future depends on the past only through the present:

$$\Pr(S_{t+1} = s' \mid S_0, \dots, S_t) = \Pr(S_{t+1} = s' \mid S_t) = P(s' \mid s).$$

The transition matrix P has rows in $\Delta(\mathcal{S})$ ($P\mathbf{1} = \mathbf{1}$).

Estimation tools (for the model-free section)

- Unbiased estimator: $\widehat{\theta}$ with $\mathbb{E}[\widehat{\theta}] = \theta$.
- Robbins–Monro step sizes: $\sum_n \alpha_n = \infty$, $\sum_n \alpha_n^2 < \infty$ drive a noisy fixed-point iteration to its target.

Used by: the Markov property defining the MDP; the induced chain P^π; the TD target as an unbiased estimate of $T^\pi V$; SARSA / Q-learning convergence.

1. The MDP model

Definition

A Markov decision process is a tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, \gamma)$:

- $\mathcal{S}, \mathcal{A}$ — finite state and action sets
- $P(s' \mid s, a)$ — transition kernel, $\sum_{s'} P(s' \mid s, a) = 1$
- $r(s, a) = \mathbb{E}[R_t \mid S_t = s, A_t = a] \in [-R_{\max}, R_{\max}]$ — expected reward
- $\gamma \in [0, 1)$ — discount factor

It generates a trajectory $S_0, A_0, R_0, S_1, A_1, R_1, \ldots$ obeying the Markov property:

$$\Pr(S_{t+1} = s' \mid S_0, A_0, \ldots, S_t, A_t) = P(s' \mid S_t, A_t).$$

The next state and expected reward depend on the past only through (S_t, A_t).

2. Policies

A policy chooses actions from observed history:

$$\text{history-dependent} \supset \text{Markov} \supset \text{stationary} \supset \text{deterministic.}$$

We write a stationary stochastic policy as $\pi(a \mid s)$, deterministic as $\pi(s)$.

A stationary policy reduces the MDP to a Markov reward process — the induced chain:

$$P^\pi(s' \mid s) = \sum_a \pi(a \mid s)\, P(s' \mid s, a), \qquad r^\pi(s) = \sum_a \pi(a \mid s)\, r(s, a).$$

Looking ahead

Some stationary deterministic policy will turn out to be optimal — restricting to stationary policies loses nothing.

3. Horizons and objectives

Three standard objectives from start state s under π:

- **Finite horizon H:** $\mathbb{E}_\pi[\sum_{t=0}^{H-1} R_t \mid S_0 = s]$ — optimal value is time-dependent V_h^*, solved by backward induction.
- **Discounted infinite horizon:** $\mathbb{E}_\pi[\sum_{t=0}^{\infty} \gamma^t R_t \mid S_0 = s]$.
- **Average reward:** $\rho^\pi(s) = \lim_{H\to\infty} \frac{1}{H}\mathbb{E}_\pi[\sum_{t=0}^{H-1} R_t \mid S_0 = s]$ — deferred.

We fix the discounted infinite-horizon objective

Discounting makes the return absolutely convergent and bounded:

$$|G_t| \le \sum_{k\ge 0} \gamma^k R_{\max} = \frac{R_{\max}}{1-\gamma}.$$

This boundedness is exactly what later makes the Bellman operators contractions.

4. Return, value, action-value

Return: $G_t = \sum_{k \geq 0} \gamma^k R_{t+k}$, with $G_t = R_t + \gamma G_{t+1}$.

Value function

$$V^\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s]$$

Action-value function

$$Q^\pi(s, a) = \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a]$$

Conditioning on the first action and transition links them:

$$V^\pi(s) = \sum_a \pi(a \mid s)\, Q^\pi(s, a), \quad Q^\pi(s, a) = r(s, a) + \gamma \sum_{s'} P(s' \mid s, a)\, V^\pi(s').$$

Optimal functions: $V^*(s) = \max_\pi V^\pi(s)$, $Q^*(s, a) = \max_\pi Q^\pi(s, a)$, with $V^*(s) = \max_a Q^*(s, a)$.

5. Bellman equations

Expectation (linear in the value)

$$V^\pi(s) = \sum_a \pi(a \mid s)\Big[r(s,a) + \gamma \sum_{s'} P(s' \mid s,a) V^\pi(s')\Big]$$

$$Q^\pi(s,a) = r(s,a) + \gamma \sum_{s'} P(s' \mid s,a) \sum_{a'} \pi(a' \mid s') Q^\pi(s',a')$$

Optimality (nonlinear: a $\max$)

$$V^*(s) = \max_a \Big[r(s,a) + \gamma \sum_{s'} P(s' \mid s,a) V^*(s')\Big]$$

$$Q^*(s,a) = r(s,a) + \gamma \sum_{s'} P(s' \mid s,a) \max_{a'} Q^*(s',a')$$

6. Bellman operators *(optional)*

View value functions as vectors in $\mathbb{R}^{\mathcal{S}}$. Define

$$(T^\pi V)(s) = r^\pi(s) + \gamma (P^\pi V)(s), \qquad (T^* V)(s) = \max_a \left[r(s,a) + \gamma \sum_{s'} P(s' \mid s,a) V(s') \right].$$

The Bellman equations become fixed-point equations:

$$\boxed{V^\pi = T^\pi V^\pi} \qquad \boxed{V^* = T^* V^*}$$

Action-value versions T_Q^π, T_Q^* on $\mathbb{R}^{\mathcal{S}\times\mathcal{A}}$ have fixed points Q^π, Q^*.

Two structural properties

Monotonicity: $V \le V' \Rightarrow T^\pi V \le T^\pi V'$ and $T^* V \le T^* V'$.
Constant shift: $T^\pi(V + c\mathbf{1}) = T^\pi V + \gamma c\mathbf{1}$ (same for T^*).

7. Contraction and uniqueness *(optional)*

Theorem (contraction)

T^π and T^* are γ-contractions in the sup-norm:

$$\|T^* V - T^* V'\|_\infty \leq \gamma \|V - V'\|_\infty.$$

Corollary (Banach fixed point)

Each operator has a unique fixed point, reached geometrically from any V_0:

$$\|V_k - V^{\text{fix}}\|_\infty \leq \gamma^k \|V_0 - V^{\text{fix}}\|_\infty.$$

Key consequence

The fixed point of T^* *is* V^*, and any policy greedy w.r.t. V^* is optimal. A single **stationary deterministic** policy is optimal for every initial state — even against all history-dependent, randomized policies.

8. Value iteration — the idea

Goal: find V^* when the model P, r is known, without solving anything by hand.

V is a vector: one number per state, $V \in \mathbb{R}^{\mathcal{S}}$ (a lookup table $V(s)$). A sweep updates *every* entry.

One backup = one step of lookahead

Start from a guess V_0 (e.g. all zeros — "I know nothing"). Repeatedly ask at each state:

"If I take the best action now and then trust my current estimate for the rest, what is this state worth?"

$$V_{k+1}(s) = \max_a \Big[\underbrace{r(s,a)}_{\text{reward now}} + \gamma \underbrace{\textstyle\sum_{s'} P(s' \mid s,a)\, V_k(s')}_{\text{discounted future, current guess}} \Big].$$

8. Value iteration — guarantees

Algorithm

Choose V_0; iterate $V_{k+1} = T^* V_k$ until a stopping rule; output the greedy policy $\pi_k = \mathcal{G}(V_k)$, where $\mathcal{G}(V)(s) \in \arg\max_a [r(s,a) + \gamma \sum_{s'} P(s' \mid s,a) V(s')]$.

Convergence $V_k \to V^*$ is immediate from the contraction. Two error bounds:

- **A-priori:** $\|V_k - V^*\|_\infty \le \gamma^k \|V_0 - V^*\|_\infty$.
- **A-posteriori (computable stopping):** $\|V_k - V^*\|_\infty \le \dfrac{\gamma}{1-\gamma} \|V_k - V_{k-1}\|_\infty$.

Greedy-policy bound

If $\|V - V^*\|_\infty \le \varepsilon$ and $\pi = \mathcal{G}(V)$, then $\|V^\pi - V^*\|_\infty \le \dfrac{2\gamma\varepsilon}{1-\gamma}$ — accurate $V \Rightarrow$ near-optimal policy.

9. Policy iteration — the idea

Goal: find an optimal policy when the model P, r is known — by improving a *policy* directly instead of polishing a value vector.

One round = evaluate, then improve

- Evaluate: take the current policy π and work out exactly how good it is at every state, V^π ("if I always follow π, what is each state worth?").
- Improve: at each state, switch to the action that looks best *given those values* — act greedily w.r.t. V^π to get a new policy π'.

9. Policy iteration — guarantees

Start from any deterministic π_0; repeat:

1. **Policy evaluation:** solve the linear system $(I - \gamma P^{\pi_k})V = r^{\pi_k}$ for V^{π_k} (invertible since $\gamma < 1$), or iterate T^{π_k}.
2. **Policy improvement:** $\pi_{k+1} = \mathcal{G}(V^{\pi_k})$, greedy w.r.t. its own value.

Stop when $\pi_{k+1} = \pi_k$.

Theorem (policy improvement)

Let $\pi' = \mathcal{G}(V^{\pi})$. Then $V^{\pi'} \geq V^{\pi}$ pointwise, with equality iff π is optimal.

Finite termination

There are finitely many ($|\mathcal{A}|^{|\mathcal{S}|}$) stationary deterministic policies and each step strictly improves until optimal — so PI terminates exactly at $V^{\pi} = V^*$.

10. Generalized policy iteration

VI and PI are the two extremes of one scheme that interleaves:

- **evaluation** — move the value estimate toward V^π (one or more T^π sweeps, or exact solve);
- **improvement** — make the policy greedy w.r.t. the current value, $\pi \leftarrow \mathcal{G}(V)$.

at any granularity.

Policy iteration	evaluation run to convergence before each improvement
Value iteration	a single greedy backup ($= T^*$)

Any convergent GPI scheme shares one fixed point: "V is the value of π" *and* "π is greedy w.r.t. V" holds iff $V = T^* V$, i.e. $V = V^*$.

11. Monte Carlo methods

First step away from a known model: learn from sampled experience. The simplest idea — average complete returns, no model and no bootstrapping.

Estimate a value by averaging returns

Since $V^\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s]$ is an expectation, estimate it by averaging the actual returns observed after visiting s over many episodes:

$$V(S_t) \leftarrow V(S_t) + \alpha\big[\, G_t - V(S_t)\,\big], \qquad G_t = \textstyle\sum_{k \geq 0} \gamma^k R_{t+k}.$$

The target is the *full* return G_t — so episodes must terminate.

Pros: unbiased; needs no model; no bootstrapping.

Cons: high variance; updates only at episode end (offline).

Control: GPI with Q estimated by averaging returns + ε-greedy (GLIE) $\rightarrow$ optimal. *Next:* TD bootstraps instead of waiting for G_t.

12. Temporal-difference methods

Instead of waiting for the full return (Monte Carlo), bootstrap: replace the expectation in a Bellman backup by a single sampled transition (S_t, A_t, R_t, S_{t+1}) and update incrementally — stochastic approximation.

TD(0) — evaluate V^π

$V(S_t) \leftarrow V(S_t) + \alpha_t\,\delta_t, \quad \delta_t = R_t + \gamma V(S_{t+1}) - V(S_t)$. Samples T^π.

SARSA — on-policy

$$Q \leftarrow Q + \alpha_t[R_t + \gamma Q(S_{t+1}, A_{t+1}) - Q]$$

Samples T_Q^π; with GLIE exploration $\to Q^*$.

Q-learning — off-policy

$$Q \leftarrow Q + \alpha_t[R_t + \gamma \max_{a'} Q(S_{t+1}, a') - Q]$$

Samples T_Q^* directly; learns Q^* under any exploratory behaviour.

Convergence (cited): Robbins–Monro step sizes $\sum_n \alpha_n = \infty$, $\sum_n \alpha_n^2 < \infty$ + all pairs visited infinitely often $\Rightarrow Q \to Q^*$ (Watkins–Dayan 1992; Tsitsiklis 1994; SARSA: Singh et al. 2000).

13. Synthesis

Each algorithm is the corresponding operator made iterative — then (bottom two rows) sampled.

Object	Defining equation	Operator	Algorithm
V^π	$V^\pi = r^\pi + \gamma P^\pi V^\pi$	T^π (linear)	policy evaluation
Q^π	Bellman exp. (Q)	T_Q^π	SARSA target
V^*	$V^* = \max_a[r + \gamma P V^*]$	T^*	value iteration
Q^*	Bellman opt. (Q)	T_Q^*	Q-learning target
opt. π	$\pi = \mathcal{G}(V^*)$	$\mathcal{G}$	policy improvement

The one-line spine

Bounded returns $\Rightarrow$ Bellman fixed-point equations $\Rightarrow$ γ-contractions $\Rightarrow$ unique fixed points reached geometrically. VI / PI / GPI iterate value-space operators with a known model; SARSA / Q-learning iterate sampled Q-space operators when the model is unknown.

Thank you